Helix Issues Workable Items

Generated from docs/workable_items.db. This document does not assert remote HelixTrack synchronization.

[ATM-019] Complete revision-pinned research for every deduplicated reader repository

Status: In progress **Type:** Task **Severity:** High **Reopens:** 0

Description: Complete revision-pinned research for every deduplicated reader repository. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: All repositories listed in Opensourced_Reader_Apps_And_Repos.md.

Reproduction / current observation: Schema-2 summary currently records 106 analyses and five unavailable checkouts; partial clones are not research evidence.

Acceptance: Every target has a complete checkout, exact revision, file ledger, manual architecture review, license review, and porting decision.

Closure criteria: Every target has a complete checkout, exact revision, file ledger, manual architecture review, license review, and porting decision.

Evidence: research/analysis_summary.json; research/reports

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-020] Perform manual architecture, code, license, and innovation review of indexed repositories

Status: Queued **Type:** Task **Severity:** High **Reopens:** 0

Description: Perform manual architecture, code, license, and innovation review of indexed repositories. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: The 106 statically indexed research checkouts and any later recovered targets.

Reproduction / current observation: Automated keyword indexing exists but does not satisfy manual architecture or license review.

Acceptance: Each repository receives an auditable manual comparison and attributed implementation decision linked to tests.

Closure criteria: Each repository receives an auditable manual comparison and attributed implementation decision linked to tests.

Evidence: research/analysis_summary.json

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-021] Build and execute Helix-specific HelixQA challenge banks

Status: In progress **Type:** Feature **Severity:** Critical **Reopens:** 0

Description: Build and execute Helix-specific HelixQA challenge banks. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: HelixQA CLI, own-org dependency modules, and Helix end-to-end challenges.

Reproduction / current observation: The actual CLI build failed because seven declared dependency checkouts are absent; only VisionEngine cloned successfully.

Acceptance: All declared dependencies resolve, Helix-specific banks run live, and machine results preserve every failure and skip.

Closure criteria: All declared dependencies resolve, Helix-specific banks run live, and machine results preserve every failure and skip.

Evidence: evidence/helixqa-build-test.log; submodules/qa/helix-deps.yaml

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-022] Deliver catalog and reader clients for native desktop, mobile, and workstation platforms

Status: Queued **Type:** Feature **Severity:** Critical **Reopens:** 0

Description: Deliver catalog and reader clients for native desktop, mobile, and workstation platforms. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: Windows, macOS, Linux, Android, iOS, and additional device classes.

Reproduction / current observation: Only the disclosed web prototype exists; no native application matrix is delivered or validated.

Acceptance: Each promised client has installable artifacts, real reader functionality, accessibility support, and device-specific live evidence.

Closure criteria: Each promised client has installable artifacts, real reader functionality, accessibility support, and device-specific live evidence.

Evidence: docs/architecture.md

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-023] Provide interactive demonstrations across the platform and device matrix

Status: Queued **Type:** Feature **Severity:** High **Reopens:** 0

Description: Provide interactive demonstrations across the platform and device matrix. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: Phone, tablet, desktop, workstation, low-end device, and varied screen demonstrations.

Reproduction / current observation: Four browser viewport checks exist but are not native-device demonstrations.

Acceptance: Interactive demos exercise real client behavior with explicit hardware and runtime evidence.

Closure criteria: Interactive demos exercise real client behavior with explicit hardware and runtime evidence.

Evidence: evidence/ui/final-run.log

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-024] Qualify containerized build, deployment, restart, and distribution workflows

Status: Queued **Type:** Feature **Severity:** Critical **Reopens:** 0

Description: Qualify containerized build, deployment, restart, and distribution workflows. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: Containers submodule, Docker or Podman, compose, and distribution providers.

Reproduction / current observation: Dockerfiles and compose configuration exist, but container tooling is unavailable and no container run has occurred.

Acceptance: Cold start, health, restart, upgrade, rollback, and distribution tests pass in actual container runtimes.

Closure criteria: Cold start, health, restart, upgrade, rollback, and distribution tests pass in actual container runtimes.

Evidence: deploy/docker-compose.yml; server/Dockerfile; submodules/containers

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-025] Configure and validate Firebase distribution without treating configuration as deployment

Status: Queued **Type:** Feature **Severity:** High **Reopens:** 0

Description: Configure and validate Firebase distribution without treating configuration as deployment. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: Firebase application distribution and associated release metadata.

Reproduction / current observation: No Firebase configuration or CLI qualification has been delivered in this tree.

Acceptance: A real project-bound preflight and distribution run produces non-secret machine evidence or an explicit skip.

Closure criteria: A real project-bound preflight and distribution run produces non-secret machine evidence or an explicit skip.

Evidence: docs/deployment.md

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-026] Validate interoperability on real readers, OPDS clients, and remote providers

Status: Queued **Type:** Feature **Severity:** Critical **Reopens:** 0

Description: Validate interoperability on real readers, OPDS clients, and remote providers. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: KOReader devices, third-party OPDS clients, and deployment-specific rclone providers.

Reproduction / current observation: Protocol-level tests exist, but no physical reader, external OPDS client, or provider-specific deployment run is recorded.

Acceptance: External clients complete real acquisition and sync workflows with captured version and environment evidence.

Closure criteria: External clients complete real acquisition and sync workflows with captured version and environment evidence.

Evidence: docs/testing.md

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-027] Remove catalog and API scalability bottlenecks under measured enterprise loads

Status: In progress **Type:** Task **Severity:** Critical **Reopens:** 0

Description: Remove catalog and API scalability bottlenecks under measured enterprise loads. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: PostgreSQL writes and search, API pagination, job retention, and connection pools.

Reproduction / current observation: Global catalog write serialization, offset pagination, per-work child-query fan-out, and in-memory job history remain measured design risks.

Acceptance: Representative large-catalog benchmarks and load tests establish limits, durable queue behavior, and bounded query plans.

Closure criteria: Representative large-catalog benchmarks and load tests establish limits, durable queue behavior, and bounded query plans.

Evidence: docs/scalability-review.md; server/internal/catalog/postgres.go; server/internal/httpapi/jobs.go

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-029] Implement verified bidirectional HelixTrack synchronization

Status: In progress **Type:** Feature **Severity:** Critical **Reopens:** 0

Description: Implement verified bidirectional HelixTrack synchronization. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: HelixTrack mapping, idempotency, conflict handling, and remote receipts.

Reproduction / current observation: Only an offline outbox contract is being established; exact remote API payloads and credentials are unverified.

Acceptance: Documented API-compatible push and pull operations pass against a real HelixTrack instance without inventing endpoints.

Closure criteria: Documented API-compatible push and pull operations pass against a real HelixTrack instance without inventing endpoints.

Evidence: docs/tracking/helixtrack-outbox.json

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-031] Implement OPDS-PSE, LCP fulfillment, and remaining external reader integrations

Status: Queued **Type:** Feature **Severity:** High **Reopens:** 0

Description: Implement OPDS-PSE, LCP fulfillment, and remaining external reader integrations. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: Page streaming, DRM fulfillment, and vendor-specific synchronization.

Reproduction / current observation: Documentation explicitly identifies these surfaces as unimplemented; detection never implies fulfillment.

Acceptance: Each integration has a working implementation, legal DRM handling, interoperability tests, and honest capability reporting.

Closure criteria: Each integration has a working implementation, legal DRM handling, interoperability tests, and honest capability reporting.

Evidence: docs/api.md; docs/faq.md; docs/formats.md

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-032] Complete rendering, conversion, and content-AI fulfillment for every required format

Status: Queued **Type:** Feature **Severity:** Critical **Reopens:** 0

Description: Complete rendering, conversion, and content-AI fulfillment for every required format. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: All formats in Additional_Formats_We_MUST_Support.jpg and their fulfillment stages.

Reproduction / current observation: Current detection and parsing coverage must not be confused with complete rendering, conversion, AI, or DRM support.

Acceptance: The per-format matrix has live tests for every claimed stage and explicit non-pass entries for the rest.

Closure criteria: The per-format matrix has live tests for every claimed stage and explicit non-pass entries for the rest.

Evidence: docs/formats.md; Additional_Formats_We_MUST_Support.jpg

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-033] Run final clean qualification and produce fresh ZIP and tar.gz archives

Status: Queued **Type:** Task **Severity:** Critical **Reopens:** 0

Description: Run final clean qualification and produce fresh ZIP and tar.gz archives. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: Clean extraction, full tests, secret scan, deterministic archive-content validation, and release notes.

Reproduction / current observation: No new-round archive is permitted while requested features, research, external validation, and qualification remain open.

Acceptance: All blocking work is genuinely closed, final evidence is refreshed, and both archives pass clean-room verification.

Closure criteria: All blocking work is genuinely closed, final evidence is refreshed, and both archives pass clean-room verification.

Evidence: docs/testing/results; CHANGELOG.md

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.